

MECKLENBURG-BRUNSWICK, VA, USA

WMO: 724118

Lat: 36.688N

Lon: 78.054W

Elev: 135

StdP: 99.72

Time Zone: -5.00 (NAE)

Period: 97-19

WBAN: 93797

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-7.8	-5.7	-17.2	0.8	-4.9	-14.0	1.1	-1.7	8.7	12.6	7.7	11.3	1.4	10	0.384

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.1	35.2	24.4	33.7	24.1	32.4	24.0	27.0	30.8	26.2	30.3	25.5	29.9	2.7	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.1	21.9	28.8	25.1	20.5	28.2	24.1	19.3	27.0	85.5	31.2	81.8	30.6	79.0	29.7	30.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
7.4	6.1	5.3	-13.0	37.0	3.7	2.3	-15.6	38.7	-17.8	40.0	-19.9	41.3	-22.6	43.0		
			-13.0	27.5	3.3	1.2	-15.4	28.4	-17.3	29.1	-19.1	29.8	-21.5	30.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	15.5	4.5	6.2	9.7	15.0	19.7	24.1	26.0	25.2	22.1	16.1	10.2	6.6
	DBStd	8.79	6.04	5.24	5.47	4.70	3.91	3.22	2.55	2.65	3.32	4.68	4.77	5.24
	HDD10.0	598	192	129	74	9	0	0	0	0	6	54	134	
	HDD18.3	1921	431	341	272	122	33	2	0	8	101	245	366	
	CDD10.0	2603	20	22	66	159	300	422	498	472	362	194	61	28
	CDD18.3	884	1	1	4	22	75	174	239	214	120	31	2	1
	CDH23.3	8319	2	3	51	241	697	1720	2394	1997	937	253	18	6
	CDH26.7	3452	0	0	7	61	223	756	1130	876	332	66	1	1
Wind	WSAvg	1.8	2.1	2.2	2.3	2.4	1.9	1.6	1.5	1.3	1.5	1.6	1.8	1.8
Precipitation	PrecAvg	1113	85	67	110	85	95	98	117	103	103	85	77	85
	PrecMax	1658	162	139	185	144	214	226	232	244	367	218	215	201
	PrecMin	828	23	26	12	28	33	35	50	41	8	0	11	18
	PrecStd	178	34	30	43	38	42	50	42	46	80	52	49	42
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	22.5	23.2	27.6	31.2	32.8	37.3	37.6	37.1	33.9	32.0	26.2	23.7
		MCWB	17.3	17.0	17.5	18.4	22.9	23.2	24.6	24.7	23.1	23.0	19.1	17.8
	2%	DB	19.1	20.9	24.1	28.0	31.1	34.0	35.3	34.1	32.3	27.9	22.8	20.3
		MCWB	16.0	14.6	16.1	18.3	22.1	23.5	24.8	24.8	23.0	21.1	17.4	16.8
	5%	DB	17.3	17.9	22.2	26.3	29.0	32.5	33.7	32.6	30.3	26.3	21.1	17.7
		MCWB	13.9	12.8	15.0	17.7	21.6	23.5	24.4	24.7	22.9	20.2	16.8	14.8
	10%	DB	14.0	15.9	19.1	23.8	27.5	31.2	32.4	31.3	28.2	23.8	18.6	15.8
		MCWB	10.3	11.4	13.6	16.7	20.3	23.2	24.4	24.4	22.1	19.2	15.3	12.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	19.1	18.7	19.2	22.1	25.0	26.6	27.9	28.1	27.1	24.9	22.0	20.0
		MCDB	20.7	21.1	24.1	26.1	29.5	30.9	32.0	32.5	28.8	27.8	23.0	21.5
	2%	WB	17.0	16.7	17.7	20.6	23.7	25.6	27.0	26.9	25.6	23.4	19.4	17.8
		MCDB	18.6	19.1	21.8	25.3	28.3	30.5	31.2	30.8	28.8	25.8	21.7	19.1
	5%	WB	14.6	14.4	16.6	19.2	22.7	25.0	26.3	26.1	24.5	22.2	17.6	15.5
		MCDB	16.7	17.4	20.4	23.9	27.2	29.9	30.7	30.1	27.4	24.9	19.2	17.7
	10%	WB	11.2	11.4	14.9	18.0	21.7	24.2	25.5	25.3	23.6	20.5	15.6	12.9
		MCDB	13.7	14.7	18.8	21.8	25.9	28.6	30.3	29.5	26.0	23.1	18.4	15.0
Mean Daily Temperature Range	5% DB	MDBR	10.8	11.6	12.5	13.0	12.1	11.7	11.1	10.9	10.9	12.0	12.1	10.6
		MCDBR	13.7	14.8	15.2	15.5	13.9	13.5	13.2	12.6	13.1	13.6	14.0	12.9
		MCWBR	10.2	10.1	8.4	7.3	6.4	5.1	4.9	4.9	5.6	7.4	9.0	9.7
	5% WB	MCDBR	12.2	13.2	13.2	12.6	11.6	11.2	10.8	10.4	9.8	10.5	11.1	11.2
		MCWBR	10.5	10.6	8.8	7.2	6.4	5.0	4.8	4.9	5.2	6.3	8.7	9.7
Clear-Sky Solar Irradiance	taub	0.298	0.312	0.343	0.386	0.432	0.478	0.508	0.510	0.401	0.355	0.317	0.300	
	taud	2.538	2.497	2.446	2.330	2.223	2.144	2.082	2.058	2.358	2.473	2.538	2.566	
	Ebn at Noon	904	929	924	896	855	813	785	774	852	866	872	877	
	Edh at Noon	78	91	105	125	141	153	162	162	114	93	77	71	
All-Sky Solar Radiation	RadAvg	2.45	3.17	4.16	5.27	5.88	6.37	6.01	5.44	4.46	3.57	2.71	2.14	
	RadStd	0.18	0.37	0.32	0.42	0.47	0.42	0.32	0.29	0.41	0.44	0.22	0.18	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (63 neighbors)	+0.40	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+144	+96

Nomenclature: See separate page